TweetBiDAF + answer reranker

Model architecture | one input question, one selected answer

1 Token representation and contextual encoding

FastText is trained only on the training partition; no external pretrained vectors.

Tweet: Tom visited Perth on Monday .

Question: Where did Tom visit ?

Same embedding layers and encoder weights for both sequences

Word embedding: 100d
FastText initialization; trainable

Raw-token Character CNN: 150d
32d chars; widths 3/4/5; 50 each

Token flags: 8d
Handcrafted indicators

Concatenate (258d) -> 2-layer Highway

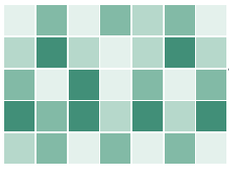
LSTM LSTM LSTM LSTM
LSTM LSTM LSTM LSTM

LSTM LSTM LSTM LSTM
LSTM LSTM LSTM LSTM

Shared 1-layer BiLSTM: H and U; 128 per direction -> 256d

2 Question-aware tweet representations

BiDAF cross-attention



S: tweet x question

C2Q = softmax(S) U

Q2C = softmax(max S) H

[H; C2Q; H*C2Q; H*Q2C]
1024d interactions
Linear + ReLU -> 256d

C2Q: attend to question tokens. Q2C: summarize question-relevant tweet tokens.

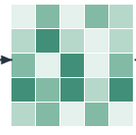
3 Gated self-attention over tweet tokens

Projected tweet states Z

Q

K

V



4 heads

Gate g
sigmoid([Z; A])

Gate + residual
LayerNorm
X = LN(Z + g*A)

A = multi-head attention(Z, Z, Z). Dropout is omitted from this schematic.

4 Additional BiLSTMs and explicit span predictions

X: question-aware tweet states

M1: 2-layer modelling BiLSTM
New layers, after attention

M2: 1 additional modelling BiLSTM
Input: M1; output: M2

Linear([X; M1]) + softmax

Linear([X; M2]) + softmax

Token	0: Tom	1: visited	2: Perth	3: on	4: Monday	5: .
p(start)	0.03	0.02	0.70	0.15	0.08	0.02
p(end)	0.02	0.02	0.38	0.48	0.08	0.02

Enumerate valid (i, j): i <= j, length <= 15. Score = p(start=i) x p(end=j).

5 Top-10 candidates -> LambdaMART reranker -> answer

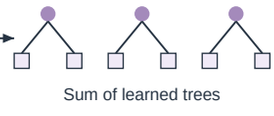
Candidate (i, j)	Base
Perth on (2, 3)	0.336
Perth (2, 2)	0.266
on (3, 3)	0.072

... 7 more candidates

10 x 30

features

LambdaMART



Sum of learned trees

Scores
0.4
1.7
-0.2

argmax Perth

Features: base score/rank, boundaries/length, overlap, question type, tweet region, flags.

One scalar score per candidate, not a class label. Reranker fitted on OOF candidates.

Illustrative probabilities, attention weights and reranker scores; gold is not an inference input.